

$$\begin{aligned}
 1 \quad D) \quad x^{(k+1)} &= x^{(k)} - \eta (8x^{(k)} + 12) \\
 x^{(k+1)} + \frac{3}{2} &= x^{(k)} + \frac{3}{2} - 8x^{(k)}\eta - 12\eta \\
 &= (1 - 8\eta) \left(x^{(k)} + \frac{3}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 z^{(k+1)} &= x^{(k+1)} + \frac{3}{2} \\
 \Rightarrow z^{(k+1)} &= (1 - 8\eta) z^{(k)} \\
 \alpha = (1 - 8\eta) \Rightarrow z^{(k+1)} &= \alpha z^{(k)}
 \end{aligned}$$

$$2 \quad A) \quad \sum_{i=0}^n (L_i - P)^2$$

$$B) \quad \nabla L = \begin{bmatrix} 2(L_1 - P) \\ 2(L_2 - P) \\ \vdots \\ 2(L_i - P) \end{bmatrix}$$

It's 0 when every one's location is P.